# Henry Hsu

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## SUMMARY

**Software Engineering Intern at Google and former GenAI Team Lead, building product features end to end: React front ends, Django/FastAPI services, and the AI systems behind them. Grew a B2B platform's users from 3K to 20K while cutting LLM token usage by 67%, and raised test coverage from zero to 67%. LlamaIndex contributor with 14 merged PRs. M.C.S. at Rice, Dec. 2026.**

## WORK EXPERIENCE

### Software Engineering Intern — Jun. 2026 – Aug. 2026

***Google*** — *Taipei, Taiwan*

- **Test Infrastructure:** Built **3** integration test infrastructures from a **zero** baseline (**Java**, **Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage:** Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling:** Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

### Generative AI Team Lead — Dec. 2024 – Aug. 2025

***MaiAgent Co., Ltd.*** *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Building AI Agents:** Upgraded chatbots into AI agents with memory systems, tool APIs, and MCP clients, driving **120%** partner growth (CTBC Bank, MSI, HPE) and user growth from **3K** to **20K** while cutting LLM token usage **67%+**
- **Real-Time Collaboration:** Built a **WebSocket** notification system on **Redis** pub/sub, plus an event-driven architecture for agent state transitions that pushes render triggers to the frontend for live UI updates
- **API Performance:** Tuned **140+** RESTful APIs through SQL refactoring, connection pooling and Django caching, cutting response time on 13 high-traffic endpoints **27.7%** and removing N+1 queries

### Generative AI Intern — Sep. 2024 – Dec. 2024

***MaiAgent Co., Ltd.*** *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Pipeline Scaling:** Refactored with an **asyncio**/**Celery** async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks
- **Reranker Integration:** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5**

## EDUCATION

### Rice University - Top 20 U.S. University — Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

### National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

## PROJECTS

**Full-Stack Social Network (Ricebook)** | *React, Redux Toolkit, Express, MongoDB, OAuth 2.0, Vitest* — **Sep. 2025 – Dec. 2025**

- **Full-Stack Build:** Built a **React 19** single-page app on Redux Toolkit against an **Express 5**/MongoDB API, with session-cookie auth plus **Google OAuth 2.0** linking, a follower graph, a comment-threaded feed, and Cloudinary image upload
- **Testing & Deployment:** Covered the Redux store with **30** Vitest suites at **97%** statement coverage and the API with Jasmine integration tests; deployed frontend and backend to separate hosts with CORS-aware session config

**Relational Database Engine from Scratch** | *C++, B+ Tree, Query Optimization, TPC-H* — **Aug. 2025 – Dec. 2025**

- **Storage & Query Engine:** Built the C++ persistence layer, a buffer manager with **LRU** eviction, a **B+ tree** index, and a **sort-merge join** over external merge sort, under a SQL compiler and cost-based optimizer; ran **TPC-H** at scale factor 1

## AWARDS

**Presidential Hackathon Winner (2024, Top 5 Nationally)** - Urban noise agent: structured-output LLM pipeline and tool calling

**23rd Golden Peak Award (2025)** - Outstanding Commercial Product, MaiAgent AI Platform

## SKILLS

**Languages:** TypeScript, Python, Java, C/C++ | **Frontend:** React, Redux, JavaScript, HTML/CSS | **Backend:** Node.js/Express, FastAPI
**Databases:** PostgreSQL, Elasticsearch, MongoDB, Milvus | **GenAI:** LlamaIndex, LangGraph, MCP | **Async:** asyncio, Celery
**Testing:** Vitest, pytest, JUnit | **APIs:** REST, WebSocket, Redis Pub/Sub | **DevOps:** GitHub Actions, Docker, git | **AWS:** EC2, S3, RDS